

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	States or uses $h = 1.5$	B1	1.1a
	Full attempt at the trapezium rule $= \frac{h}{2} \{1.63 + 2.63 + 2 \times (2 + 2.26 + 2.46)\}$	M1	1.1b
	= awrt 13.3 or $\frac{531}{40}$	A1	1.1b
		(3)	
(b)(i)	$\int_3^9 \log_3(2x)^{10} dx = 10 \times "13.3" = \text{awrt } 133 \text{ or e.g. } \frac{531}{4}$	B1ft	2.2a
(ii)	$\int_3^9 \log_3 18x dx = \int_3^9 \log_3(9 \times 2x) dx = \int_3^9 2 + \log_3 2x dx$ $= [2x]_3^9 + \int_3^9 \log_3 2x dx = 18 - 6 + \int_3^9 \log_3 2x dx = \dots$	M1	3.1a
	Awrt 25.3 or $\frac{1011}{40}$	A1ft	1.1b
		(3)	
(6 marks)			
Notes:			

(a)

B1: States or uses $h = 1.5$

M1: A full attempt at the trapezium rule.

Look for $\frac{\text{their } h}{2} \{1.63 + 2.63 + 2 \times (2 + 2.26 + 2.46)\}$ but condone copying slips

Note that $\frac{\text{their } h}{2} \{1.63 + 2.63 + 2 \times (2 + 2.26 + 2.46)\}$ scores M0 unless the missing brackets are recovered or implied by their answer. You may need to check.

Allow this mark if they add the areas of individual trapezia e.g.

$$\frac{\text{their } h}{2} \{1.63 + 2\} + \frac{\text{their } h}{2} \{2 + 2.26\} + \frac{\text{their } h}{2} \{2.26 + 2.46\} + \frac{\text{their } h}{2} \{2.46 + 2.63\}$$

Condone copying slips but must be a complete method using all the trapezia.

A1: awrt 13.3 (Note full accuracy is 13.275) or exact equivalent.

Note that the calculator answer is 13.324 so you must see correct working to award awrt 13.3

Use of $h = -1.5$ leading to a negative area can score B1M1A0 but allow full marks if then stated as positive.

(b)(i)

B1ft: Deduces that $\int_3^9 \log_3(2x)^{10} dx = 10 \times "13.3" = \text{awrt } 133$

FT on their 13.3 look for 3sf accuracy but follow through on e.g. their rounded answer to part (a) so if 13 was their answer to part (a) then allow 130 here following a correct method.

A correct method must be seen here but a minimum is e.g. $10 \times "13.3" = "133"$

Note that $\int_3^9 \log_3(2x)^{10} dx = 133.2414316\dots$ so a correct method must be seen to award marks.

Attempts to apply the trapezium rule again in any way score M0 as the instruction in the question was to use the answer to part (a).

(b)(ii)

M1: Shows correct log work to relate the given question to part (a)

Must reach as far as e.g. $[2x]_3^9 + \int_3^9 \log_3 2x \, dx = \dots$ with correct use of limits on $[2x]_3^9$ which may be implied or equivalent work e.g. finds the area of the rectangle as 2×6

Alft: Correct working followed by awrt 25.3 but ft on their 13.3 so allow for $12 +$ their answer to part (a) following correct work as shown.

Note that $\int_3^9 \log_3 18x \, dx = 25.32414\dots$ so a correct method must be seen to award marks.

Some examples of an acceptable method are:

$$\int_3^9 \log_3 18x \, dx = \int_3^9 \log_3 (9 \times 2x) \, dx = \int_3^9 2 + \log_3 2x \, dx = 6 \times 2 + "13.3" = 25.3$$

$$\int_3^9 \log_3 18x \, dx = \int_3^9 \log_3 (9 \times 2x) \, dx = \int_3^9 2 + \log_3 2x \, dx = 12 + "13.3" = 25.3$$

$$\int_3^9 \log_3 18x \, dx = \int_3^9 \log_3 (9 \times 2x) \, dx = \int_3^9 2 + \log_3 2x \, dx = [2x]_3^9 + \int_3^9 \log_3 2x \, dx = 25.3$$

BUT just $12 + "13.3" = 25.3$ scores **M0**

Attempts to apply the trapezium rule again in any way score **M0** as the instruction in the question was to use the answer to part (a).

Q2.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$y = x^x \Rightarrow \ln y = x \ln x$	M1	This mark is for a method to find the x -coordinate of the turning point of C by taking logarithms
	$\ln y = x \ln x \Rightarrow \frac{1}{y} \frac{dy}{dx} = \ln x + 1$	M1	This mark is given for a method using implicit differentiation
		A1	This mark is given for a correct expression for $\frac{1}{y} \frac{dy}{dx}$
	Setting $\frac{dy}{dx} = 0$, $\ln x + 1 = 0$	M1	This mark is given for a method for finding the turning point of C by setting $\frac{dy}{dx} = 0$
	$x = e^{-1}$	A1	This mark is given for correctly finding a value for the x -coordinate of the turning point of C
(b)	$1.5^{1.5} = 1.837\dots$, $1.6^{1.6} = 2.121\dots$	M1	This mark is given for substituting 1.5 and 1.6 into $y = x^x$
	The curve C contains the points (1.5, 1.8) and (1.6, 2.1). At P , $y = 2$ Since C is continuous, $1.5 < \alpha < 1.6$	A1	This mark is given for a valid explanation that C contains the points (1.5, 1.8) and (1.6, 2.1) and is continuous
(c)	$x_1 = 1.5$ $x_2 = 2 \times 1.5^{-0.5} = 1.633$	M1	This mark is given for finding a correct value for x_2
	$x_3 = 2 \times 1.633^{-0.633} = 1.466$ $x_4 = 2 \times 1.466^{-0.466} = 1.673$	A1	This mark is given for finding a correct value for x_4
(d)	For example: x_n oscillates is periodic is non-convergent	B1	This mark is given for a valid statement about the long-term behaviour of x_n
	between 1 and 2	B1	This mark is given for stating that the behaviour is between 1 and 2
(Total 11 marks)			

Q3.

Question	Scheme	Marks	AOs
(a)	$h = 0.5$	B1	1.1b
	$A \approx \frac{1}{2} \times \frac{1}{2} \{0.4805 + 1.9218 + 2(0.8396 + 1.2069 + 1.5694)\}$	M1	1.1b
	$= 2.41$	A1	1.1b
		(3)	
(b)	$\int (\ln x)^2 dx = x(\ln x)^2 - \int x \times \frac{2 \ln x}{x} dx$	M1 A1	3.1a 1.1b
	$= x(\ln x)^2 - 2 \int \ln x dx = x(\ln x)^2 - 2(x \ln x - \int dx)$ $= x(\ln x)^2 - 2 \int \ln x dx = x(\ln x)^2 - 2x \ln x + 2x$	dM1	2.1
	$\int_2^4 (\ln x)^2 dx = [x(\ln x)^2 - 2x \ln x + 2x]_2^4$ $= 4(\ln 4)^2 - 2 \times 4 \ln 4 + 2 \times 4 - (2(\ln 2)^2 - 2 \times 2 \ln 2 + 2 \times 2)$ $= 4(2 \ln 2)^2 - 16 \ln 2 + 8 - 2(\ln 2)^2 + 4 \ln 2 - 4$	ddM1	2.1
	$= 14(\ln 2)^2 - 12 \ln 2 + 4$	A1	1.1b
		(5)	
(8 marks)			
Notes			

(a)

B1: Correct strip width. May be implied by $\frac{1}{2} \times \frac{1}{2} \{ \dots \}$ or $\frac{1}{4} \times \{ \dots \}$

M1: Correct application of the trapezium rule.

Look for $\frac{1}{2} \times "h" \{ 0.4805 + 1.9218 + 2(0.8396 + 1.2069 + 1.5694) \}$ condoning slips in the digits.

The bracketing must be correct but it is implied by awrt 2.41

A1: 2.41 only. This is not awrt

(b)

M1: Attempts parts the correct way round to achieve $\alpha x (\ln x)^2 - \beta \int \ln x \, dx$ o.e.

May be unsimplified (see scheme).

Watch for candidates who know or learn $\int \ln x \, dx = x \ln x - x$

who may write $\int (\ln x)^2 \, dx = \int (\ln x)(\ln x) \, dx = \ln x(x \ln x - x) - \int \frac{x \ln x - x}{x} \, dx$

A1: Correct expression which may be unsimplified

dM1: Attempts parts again to (only condone coefficient errors) to

achieve $\alpha x (\ln x)^2 - \beta x \ln x \pm \gamma x$ o.e.

ddM1: Applies the limits 4 and 2 to an expression of the form $\pm \alpha x (\ln x)^2 \pm \beta x \ln x \pm \gamma x$, subtracts and applies $\ln 4 = 2 \ln 2$ at least once. Both M's must have been awarded

A1: Correct answer

It is possible to do $\int (\ln x)^2 \, dx$ via a substitution $u = \ln x$ but it is very similar.

M1 A1, dM1: $\int u^2 e^u \, du = u^2 e^u - \int 2u e^u \, du = u^2 e^u - 2ue^u \pm 2e^u$

ddM1: Applies appropriate limits and uses $\ln 4 = 2 \ln 2$ at least once to an expression of the form $u^2 e^u - \beta u e^u \pm \gamma e^u$ Both M's must have been awarded

Q4.

Question	Scheme	Marks	AOs
(a)	$h = 0.5$	B1	1.1a
	$A \approx \frac{0.5}{2} \{0.5774 + 0.8452 + 2(0.7071 + 0.7746 + 0.8165)\}$	M1	1.1b
	= awrt 1.50	A1	1.1b
	For reference: The integration on a calculator gives 1.511549071 The full accuracy for y values gives 1.504726147 The accuracy from the table gives 1.50475		
		(3)	
(b)	$3 \times \text{their (a)}$ If (a) is correct, allow awrt 4.50 or awrt 4.51 even with no working. Only allow 4.5 if (a) is correct and working is shown e.g. 3×1.5 If (a) is incorrect allow $3 \times \text{their (a)}$ given to at least 3sf but do not be too concerned about the accuracy (as they may use rounded or rounded value from (a))	B1ft	2.2a
	For reference the integration on a calculator gives 4.534647213		
		(1)	
(c)	<u>This mark depends on the B1 having been awarded in part (b) with awrt 4.5</u> Look for a sensible comment. Some examples: - The answer is accurate to 2 sf or one decimal place - Answer to (b) is accurate as $4.535 \approx 4.50$ - Very accurate as 4.535 to 2 sf is 4.5 $4.51425 < 4.535$ so my answer is underestimate but not too far off - It is an underestimate but quite close - It is a very good estimate - High accuracy (Quite) accurate - It is less than 1% out $4.535 - 4.5 = 0.035$ so not far out But not just "it is an underestimate" or Calculates percentage error correctly using awrt 4.50 or awrt 4.51 or 4.5 (No comment is necessary in these cases although one may be given) Examples: $\left \frac{4.535 - 4.50}{4.535} \right \times 100 = 0.77\% \quad \text{or} \quad \left \frac{4.535 - 4.51}{4.535} \right \times 100 = 0.55\% \quad \text{or}$ $\left \frac{4.535 - 4.51425}{4.535} \right \times 100 = 0.46\% \quad \text{or} \quad \left \frac{4.50}{4.535} \right \times 100 = 99\%$ In these cases don't be too concerned about accuracy e.g. allow 1sf. This mark should be withheld if there are any contradictory statements	B1	3.2b
		(1)	
(5 marks)			

Notes:

(a)

B1: States or uses $h = 0.5$. May be implied by $\frac{1}{4} \times \{ \dots \}$ below.

M1: Correct attempt at the trapezium rule.

Look for $\frac{1}{2}h \times \{0.5774 + 0.8452 + 2(0.7071 + 0.7746 + 0.8165)\}$ condoning slips on the terms but must use all y values with no repeats.

There must be a clear attempt at $\frac{1}{2}h \times (\text{first } y + \text{last } y + 2 \times \text{"sum of the rest"})$

Give M0 for $\frac{1}{2} \times \frac{1}{2} \times 0.5774 + 0.8452 + 2(0.7071 + 0.7746 + 0.8165)$ unless the missing brackets are implied.

NB this incorrect method gives 5.85...

May be awarded for separate trapezia e.g.

$$\frac{1}{4}(0.5774 + 0.7071) + \frac{1}{4}(0.7071 + 0.7746) + \frac{1}{4}(0.7746 + 0.8165) + \frac{1}{4}(0.8165 + 0.8452)$$

May be awarded for using the function e.g. $\frac{1}{2}h \times \left\{ \sqrt{\frac{0.5}{1+0.5}} + \sqrt{\frac{2.5}{1+2.5}} + 2 \left(\sqrt{\frac{1}{1+1}} + \sqrt{\frac{1.5}{1+1.5}} + \sqrt{\frac{2}{1+2}} \right) \right\}$

A1: Awrt 1.50 (Apply isw if necessary)

Correct answers with no working – send to review

(b)

B1ft: See main scheme. Must be considering $3 \times$ (a) and not e.g. attempting trapezium rule again.

(c)

B1: See scheme

Q5.

Question	Scheme	Marks	AOs
	The equation $2x^3 + x^2 - 1 = 0$ has exactly one real root		
(a)	$\{f(x) = 2x^3 + x^2 - 1 \Rightarrow f'(x) = 6x^2 + 2x\}$	B1	1.1b
	$\left\{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \Rightarrow \left\{x_{n+1}\right\} = x_n - \frac{2x_n^3 + x_n^2 - 1}{6x_n^2 + 2x_n}\right.$	M1	1.1b
	$= \frac{x_n(6x_n^2 + 2x_n) - (2x_n^3 + x_n^2 - 1)}{6x_n^2 + 2x_n} \Rightarrow x_{n+1} = \frac{4x_n^3 + x_n^2 + 1}{6x_n^2 + 2x_n} *$	A1*	2.1
		(3)	
(b)	$\{x_1 = 1 \Rightarrow x_2 = \frac{4(1)^3 + (1)^2 + 1}{6(1)^2 + 2(1)} \text{ or } x_2 = 1 - \frac{2(1)^3 + (1)^2 - 1}{6(1)^2 + 2(1)}\}$	M1	1.1b
	$\Rightarrow x_2 = \frac{3}{4}, x_3 = \frac{2}{3}$	A1	1.1b
		(2)	
(c)	Accept any reasons why the Newton-Raphson method cannot be used with $x_1 = 0$ which refer or <i>allude</i> to either the stationary point or the tangent. E.g. • There is a stationary point at $x = 0$ • Tangent to the curve (or $y = 2x^3 + x^2 - 1$) would not meet the x -axis • Tangent to the curve (or $y = 2x^3 + x^2 - 1$) is horizontal	B1	2.3
		(1)	
(6 marks)			

Notes for Question	
(a)	
B1:	States that $f'(x) = 6x^2 + 2x$ or states that $f'(x_n) = 6x_n^2 + 2x_n$ (Condone $\frac{dy}{dx} = 6x^2 + 2x$)
M1:	Substitutes $f(x_n) = 2x_n^3 + x_n^2 - 1$ and their $f'(x_n)$ into $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
A1*:	A correct intermediate step of making a common denominator which leads to the given answer
Note:	Allow B1 if $f'(x) = 6x^2 + 2x$ is applied as $f'(x_n)$ (or $f'(x)$) in the NR formula $\{x_{n+1}\} = x_n - \frac{f(x_n)}{f'(x_n)}$
Note:	Allow M1A1 for $\bullet \quad x_{n+1} = x - \frac{2x^3 + x^2 - 1}{6x^2 + 2x} = \frac{x(6x^2 + 2x) - (2x^3 + x^2 - 1)}{6x^2 + 2x} \Rightarrow x_{n+1} = \frac{4x_n^3 + x_n^2 + 1}{6x_n^2 + 2x_n}$
Note	Condone $x = x - \frac{2x^3 + x^2 - 1}{6x^2 + 2x}$ for M1
Note	Condone $x_n - \frac{2x_n^3 + x_n^2 - 1}{6x_n^2 + 2x_n}$ or $x - \frac{2x^3 + x^2 - 1}{6x^2 + 2x}$ (i.e. no $x_{n+1} = \dots$) for M1
Note:	Give M0 for $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ followed by $x_{n+1} = 2x_n^3 + x_n^2 - 1 - \frac{2x_n^3 + x_n^2 - 1}{6x_n^2 + 2x_n}$
Note:	Correct notation, i.e. x_{n+1} and x_n must be seen in their final answer for A1*
(b)	
M1:	An attempt to use the given or their formula once. Can be implied by $\frac{4(1)^3 + (1)^2 + 1}{6(1)^2 + 2(1)}$ or 0.75 o.e.
Note:	Allow one slip in substituting $x_1 = 1$
A1:	$x_2 = \frac{3}{4}$ and $x_3 = \frac{2}{3}$
Note:	Condone $x_2 = \frac{3}{4}$ and $x_3 = \text{awrt } 0.667$ for A1
Note:	Condone $\frac{3}{4}, \frac{2}{3}$ listed in a correct order ignoring subscripts
(c)	
B1:	See scheme
Note:	Give B0 for the following isolated reasons: e.g. - You cannot divide by 0 - The fraction (or the NR formula) is undefined at $x = 0$ - At $x = 0$, $f'(x_1) = 0$ x_1 cannot be 0 $6x^2 + 2x$ cannot be 0 - the denominator is 0 which cannot happen - if $x_1 = 0$, $6x^2 + 2x = 0$

Q6.

Question	Scheme	Marks	AOs
(a)	Uses or implies $h = 0.5$	B1	1.1b
	For correct form of the trapezium rule =	M1	1.1b
	$\frac{0.5}{2} \{3 + 2.2958 + 2(2.3041 + 1.9242 + 1.9089)\} = 4.393$	A1	1.1b
		(3)	
(b)	Any valid statement reason, for example • Increase the number of strips • Decrease the width of the strips • Use more trapezia	B1	2.4
		(1)	
(c)	For integration by parts on $\int x^2 \ln x \, dx$	M1	2.1
	$= \frac{x^3}{3} \ln x - \int \frac{x^2}{3} \, dx$	A1	1.1b
	$\int -2x + 5 \, dx = -x^2 + 5x \quad (+c)$	B1	1.1b
	All integration attempted and limits used Area of $S = \int_1^3 \frac{x^2 \ln x}{3} - 2x + 5 \, dx = \left[\frac{x^3}{9} \ln x - \frac{x^3}{27} - x^2 + 5x \right]_{x=1}^{x=3}$	M1	2.1
	Uses correct ln laws, simplifies and writes in required form	M1	2.1
	Area of $S = \frac{28}{27} + \ln 27 \quad (a = 28, b = 27, c = 27)$	A1	1.1b
		(6)	
(10 marks)			

Notes:

(a)

B1: States or uses the strip width $h = 0.5$. This can be implied by the sight of $\frac{0.5}{2}\{\dots\}$ in the trapezium rule

M1: For the correct form of the bracket in the trapezium rule. Must be y values rather than x values $\{\text{first } y \text{ value} + \text{last } y \text{ value} + 2 \times (\text{sum of other } y \text{ values})\}$

A1: 4.393

(b)

B1: See scheme

(c)

M1: Uses integration by parts the right way around.

Look for $\int x^2 \ln x \, dx = Ax^3 \ln x - \int Bx^2 \, dx$

A1: $\frac{x^3}{3} \ln x - \int \frac{x^2}{3} \, dx$

B1: Integrates the $-2x + 5$ term correctly $= -x^2 + 5x$

M1: All integration completed and limits used

M1: Simplifies using $\ln$ law(s) to a form $\frac{a}{b} + \ln c$

A1: Correct answer only $\frac{28}{27} + \ln 27$